

Anchoring decisions under uncertainty

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conjoint work with
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Deciding under uncertainty

Anchor-robustness

- Project scheduling

- Deciding under uncertainty

- Anchor-Reoptimization

Anchor-Robust project scheduling

Deterministic combinatorial optimization problem

Optimization problem (Π)

Given

a set of elements E
a collection $\mathcal{Feas} \subseteq 2^E$
a cost function $w \in \mathbb{R}_+^E$

find $\min w(S)$
s.t. $S \in \mathcal{Feas}$

Deciding under uncertainty

In practice, it is never the case the data are known in advance because either

- data are estimated, e.g. from a statistical analysis
- or data will be known once the solution has been implemented
- or data will be changed several times

The questions raised by uncertainty

Three questions is raised from uncertainty

- ▶ **Reoptimization** Given a solution of an instance (called baseline instance), and a new instance, find a new solution of the new instance that is close to the baseline instance.
- ▶ **Dimensioning** What are the maximum cost, volume, size... that can happen in case of uncertainty?
 - ▶ **static-robustness** Finding a solution that is always valid (and then very expensive)
 - ▶ **adjustable-robustness** Dimensioning so that there is always a solution (and then decisions will be often ajusted)
- ▶ **Finding a robust baseline solution** Determining a baseline instance that have good properties whatever the uncertain realization
 - ▶ **static-robustness** A solution that is always valid (and then very expensive)
 - ▶ **anchor-robustness** A solution that can be easily adjust to correspond the new instance

Stochastic optimization literature

In the stochastic (or expected approach), the decision maker tries to minimize the max values obtained through an expectation measure

$$\min_{S \in Feas} E[w(S)]$$

expected function costs w depends on stochastic scenario as if “Nature played stochastically”

Robust optimization literature

Another approach (more deterministic) is to fight against a worst case

Static-Robust

$$\begin{array}{ll} \min & \max_{\delta \in \Delta} w^\delta(S) \\ \text{s.t.} & S \in \mathcal{Feas}^\delta \quad \forall \delta \in \Delta \end{array}$$

one solution S

- ✓ x is a baseline solution
- ✓ x always remains feasible
- ✗ can be very expensive

Adjustable-Robust

$$\begin{array}{ll} \max_{\delta \in \Delta} & \min w^\delta S^\delta \\ & \text{s.t. } S^\delta \in \mathcal{Feas}^\delta \end{array}$$

one solution S^δ per $\delta \in \Delta$

- ✗ no baseline solution
- ✗ decisions must adapt to δ
- ✓ lower optimal value

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Anchoring

- ▶ The decision maker is often **change-averse**
- ▶ Prefers to **maintain decisions taken previously**
= **anchor** them
- ▶ **Anchor-reoptimization problem** :
find a solution while sticking to a baseline solution
- ▶ Also (but not in this talk) **Anchor-Robust problem** :
how to incorporate the anchoring criterion in the choice of the baseline solution

Project scheduling with uncertain processing times

Data :

- ▶ Jobs $J = \{1, \dots, n\}$ with processing time p_i , $i \in J$
- ▶ Set A of precedence constraints $i \rightarrow j$
(i.e. job i must end before j starts)
with (J, A) acyclic

The Project Scheduling problem :

- ▶ **Schedule** each job i , i.e. decide **starting time** $x_i \geq 0$
- ▶ s.t. $x_i + p_i \leq x_j \quad \forall i \rightarrow j \in A$
- ▶ Minimizing the **makespan**, i.e. the total duration
$$M = \min \max_{i \in J} (x_i + p_i)$$

Example :

Instance :

8 jobs

Processing time :

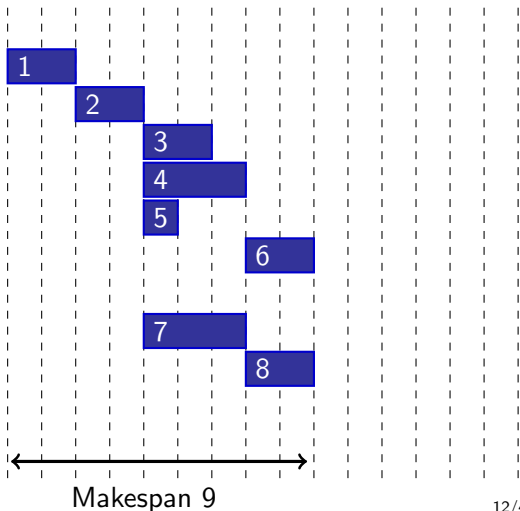
	1	2	3	4	5	6	7	8
p_i	2	2	2	3	1	2	3	2

Precedence constraints :

$1 \rightarrow 2 \rightarrow \{3, 4, 5\} \rightarrow 6$

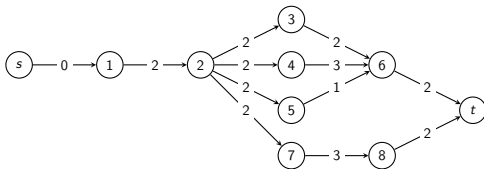
$2 \rightarrow 7 \rightarrow 8$

A solution schedule :



Two (trivial) polynomial methods

- The longest path problem in the weighted precedence graph $G(p)$
 $G(p) = (J \cup \{s, t\}, A' = A \cup \{(s \rightarrow i), (i \rightarrow t)\})$



- Solving the following linear program

$$\begin{aligned} \min \quad & x_t \\ & x_i + p_i \leq x_j \quad \forall (i \rightarrow j) \in A' \\ & x_i \geq 0 \quad \forall i \in J \cup \{s, t\} \end{aligned}$$

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Planning problems under uncertainty

Practical use cases :

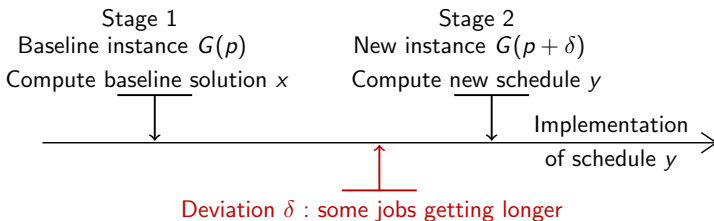
- ▶ construction (building, power plant...)
- ▶ high-tech project (aircraft carrier, new software)
- ▶ industrial maintenance

Planning problems under uncertainty

Practical use cases :

- ▶ construction (building, power plant...)
- ▶ high-tech project (aircraft carrier, new software)
- ▶ industrial maintenance

**A “two stage” decision process
in an industrial context where a baseline solution is required :**



Uncertainty :

- **uncertain data**
 - schedules are determined months in advance
 - delays in operations may occur
- **"change-averse" decision makers**
 - preparation is necessary for staff and equipment
 - better to avoid changes in schedules

Uncertainty :

- **uncertain data**

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- **"change-averse" decision makers**

- preparation is necessary for staff and equipment
- better to avoid changes in schedules

⇒ Robust problem where some decisions have to be **anchored**,
i.e. "guaranteed not to change"

⇒ Aim to favor unchanged decisions
between the baseline solutions and the new solution

Two issues :

- **Reoptimization** : how to modify the solution after the instance has changed
- **Robust optimization** : how to compute a baseline solution in advance considering uncertainty

Many optimization problems enter into this uncertainty context

- ▶ More realistic planning problem (NP-hard in general)
 - ▶ with limited resources
 - ▶ with setup production costs or delivery delay
 - ▶ public transportation
(timetabling, reoptimization after accident)

Many optimization problems enter into this uncertainty context

- ▶ More realistic planning problem (NP-hard in general)
 - ▶ with limited resources
 - ▶ with setup production costs or delivery delay
 - ▶ public transportation
(timetabling, reoptimization after accident)
- ▶ Network design
 - ▶ minimum cost connected network (spanning tree)
 - ▶ reliable network (having spare roads in case of failure)

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- ▶ Network design
 - ▶ minimum cost connected network (spanning tree)
 - ▶ reliable network (having spare roads in case of failure)
- ▶ Network routing and resource assignment (e.g. 5G slicing)
 - ▶ Rerouting to take account of increased bandwidth requests
 - ▶ Robust routing

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Reoptimization : the second-stage problem

► Inputs :

- a new instance $G(p')$
- a baseline solution x

$\rightsquigarrow$ find new solution y of instance $G(p')$

► **Anchoring level** : $\sigma(x, y) = \#$ identical decisions in x and y

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► **Anchor-Reoptimization problems** :

$$\begin{array}{ll} \max & \sigma(x, y) \\ \text{s.t.} & y \text{ schedule of } G(p') \end{array} \quad \left| \right.$$

Reoptimization : the second-stage problem

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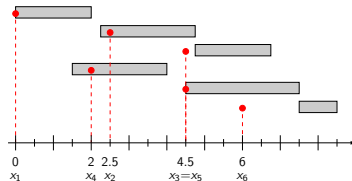
$$\begin{array}{ll} \min & c^T y \\ \text{s.t.} & y \text{ schedule of } G(p') \\ & \sigma(x, y) \geq k \end{array}$$

Rescheduling

► Inputs :

- new instance $G(p')$
- baseline schedule x : ●

↪ find new solution y

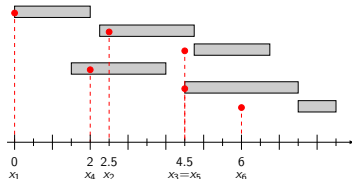


Rescheduling

► Inputs :

- new instance $G(p')$
- baseline schedule x : ●

↪ find new solution y



► **Anchoring level** : $\sigma(x, y) = |\{j \in J : x_j = y_j\}|$

► **Anchor-Rescheduling problem** :

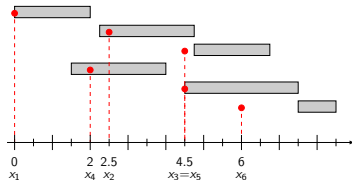
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Rescheduling

► Inputs :

- new instance $G(p')$
- baseline schedule x : ●

↪ find new solution y



► **ε -Anchoring level** : $\sigma_\varepsilon(x, y) = |\{j \in J : |x_j - y_j| \leq \varepsilon_j\}|$
for some tolerance ε .

► **ε -Anchor-Rescheduling problem** :

$$\begin{array}{ll} \max & \sigma_\varepsilon(x, y) \\ \text{s.t.} & y \text{ schedule of } G(p') \end{array}$$

An important **combinatorial property** :

Proposition

Let H be a subset of jobs.

There exists a schedule y of $G(p)$ such that $x_i = y_i \forall i \in H$

$\Leftrightarrow$

$$x_j - x_i \geq \underbrace{L_{G(p)}(i, j)}_{\text{longest } i\text{-}j \text{ path in } G(p)} \quad \forall i, j \in H$$

Polynomiality of Anchor-Rescheduling

Theorem

ϵ -Anchor-Rescheduling is solvable in polynomial time.

- ▶ Shown equivalent to a max antichain problem in appropriate poset
- ▶ Even if there are some **deadline constraints** (circuits in G)

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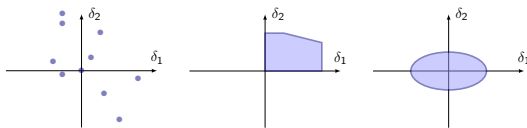
Anchor-Robust project scheduling

Robust optimization : the first-stage problem

► **Processing time uncertainty :**

nominal value $p \rightsquigarrow$ real value $p + \delta$ with $\delta \in \Delta$

► Assume deviation $\delta \in \Delta$: the **uncertainty set**



► Optimize a **worst-case criterion**.

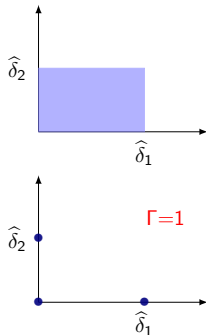
Uncertainty sets

Box $\Delta = \prod_{j \in J} [0, \hat{\delta}_j]$
 proc. time of j may be up to $p_j + \hat{\delta}_j$

Budgeted $\Delta = \{(\hat{\delta}_j u_j) : u \in \{0, 1\}^J, \sum_j u_j \leq \Gamma\}$

[Bertsimas Sim, 04]

proc. time of j is in $\{p_j, p_j + \hat{\delta}_j\}$
 and at most Γ jobs disrupted at a time



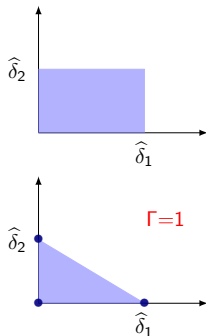
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[Bertsimas Sim, 04] proc. time of j is in $\{p_j, p_j + \hat{\delta}_j\}$
 and at most Γ jobs disrupted at a time

► Δ can be convexified w.l.o.g.



Robust optimization literature

Static-Robust

$$\begin{array}{ll} \min & \max_{\delta \in \Delta} c^{\delta T} x \\ \text{s.t.} & x \in \mathcal{X}^{\delta} \quad \forall \delta \in \Delta \end{array}$$

one solution x

- ✓ x is a baseline solution
- ✓ x always remains feasible
- ✗ can be very expensive

Adjustable-Robust

$$\begin{array}{ll} \max_{\delta \in \Delta} & \min c^{\delta T} x^{\delta} \\ & \text{s.t. } x^{\delta} \in \mathcal{X}^{\delta} \end{array}$$

one solution x^{δ} per $\delta \in \Delta$

- ✗ no baseline solution
- ✗ decisions must adapt to δ
- ✓ lower optimal value

Anchor-Robustness

► New concept : Anchor-Robustness as a middle ground.

Static-Robust

Anchor-Robust

Adjustable-Robust

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one solution x

$$\begin{array}{ll} \max_{\delta \in \Delta} & \min c^{\delta T} x^{\delta} \\ & \text{s.t. } x^{\delta} \in \mathcal{X}^{\delta} \end{array}$$

one solution x^{δ} per $\delta \in \Delta$

✓ x is a baseline solution

✓ x always remains feasible

✗ can be very expensive

✓ with a baseline solution

✓ adapt only some decisions

✓ control solution cost

✗ no baseline solution

✗ decisions must adapt to δ

✓ lower optimal value

Anchor-Robust Project Scheduling Problem (AnchRobPSP)

► Inputs :

- Baseline instance : precedence graph $G(p)$
- Uncertainty set Δ
- A deadline $M \geq 0$

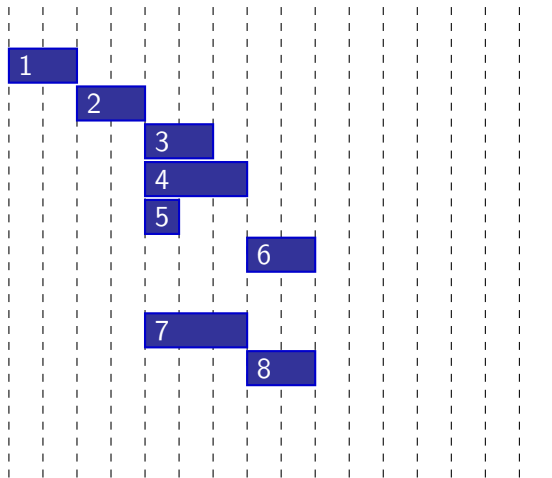
► Solution (x^0, H) :

- **A baseline schedule** x^0 feasible for $G(p)$ with $x_t^0 \leq M$
 - **A subset of anchored jobs H** : jobs with guaranteed starting times
- $\forall \delta \in \Delta$, there exists x^δ schedule of $G(p+\delta)$ such that $x_i^0 = x_i^\delta$ for every $i \in H$

► Objective :

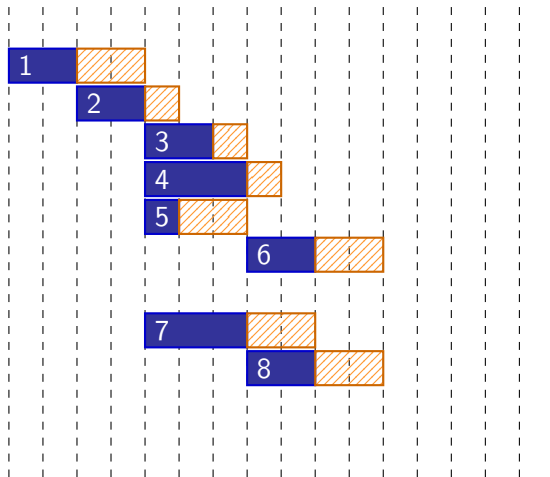
- Given weights $w \in \mathbb{R}_+^J$, maximize the total **anchoring weight** $\sum_{i \in H} w_i$

Example : (anchor-)robust problems



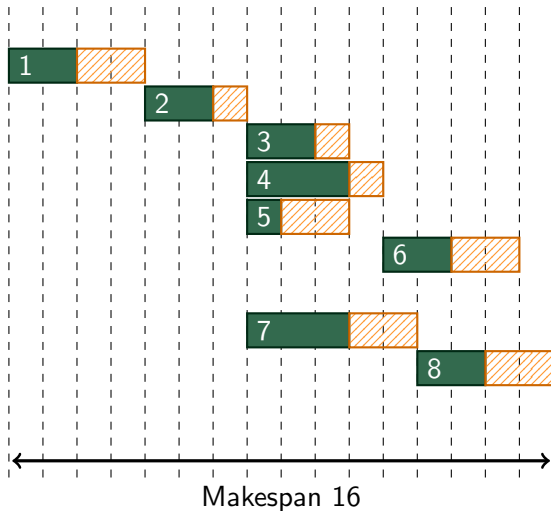
► 8 jobs with baseline processing times

Example : (anchor-)robust problems



- ▶ 8 jobs with baseline processing times
- ▶ Δ : Budgeted set with $\Gamma = 1$
i.e. each job can have increased proc. time
+ at most one proc. time will increase

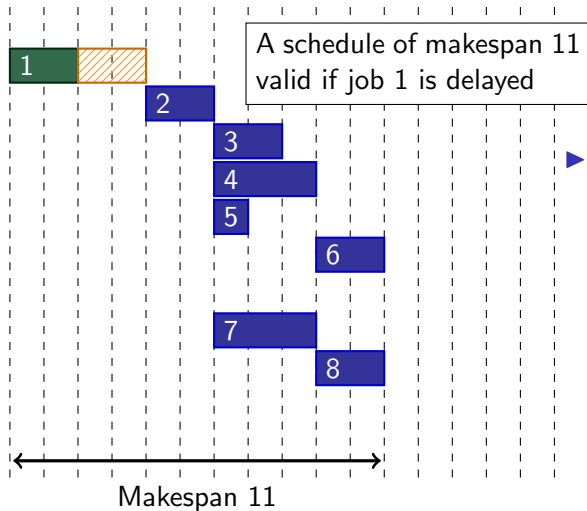
Example : (anchor-)robust problems



► Robust-static schedule :
feasible $\forall \delta \in \Delta$

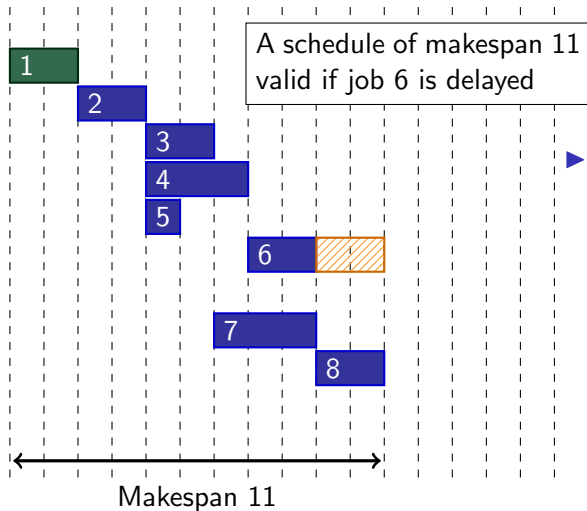
→ **all jobs anchored**
but large makespan

Example : (anchor-)robust problems



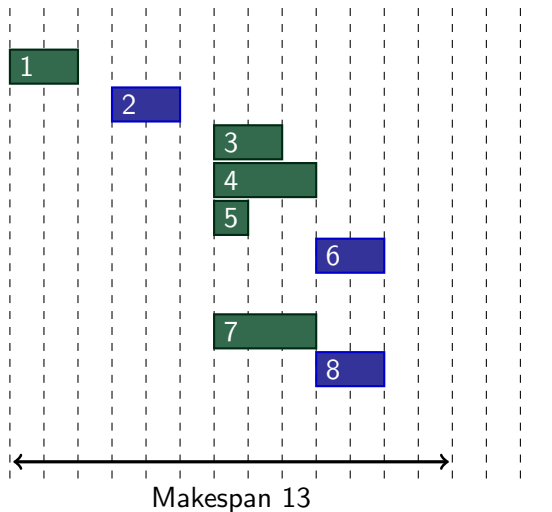
- Robust 2-stage :
[Minoux,07]
min makespan
achievable $\forall \delta \in \Delta$
→ then **only job 1**
has the same starting
time $\forall \delta \in \Delta$

Example : (anchor-)robust problems



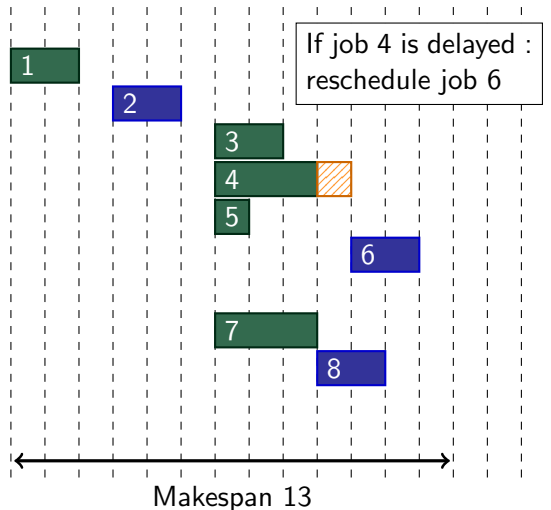
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Example : (anchor-)robust problems



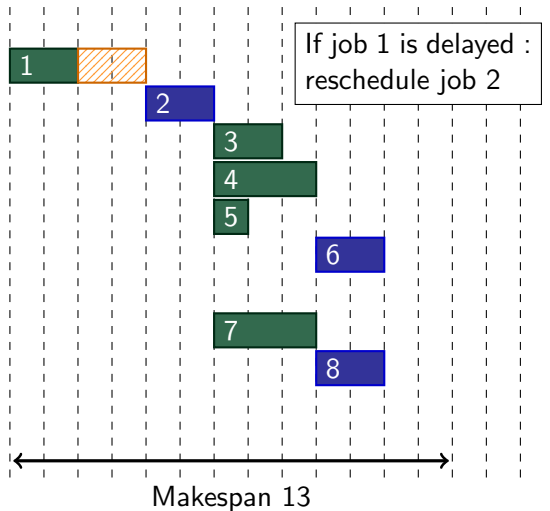
- ▶ With (AnchRob) :
for deadline $M = 13$,
we can guarantee the
starting times of
5 jobs (anchored)

Example : (anchor-)robust problems



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Example : (anchor-)robust problems



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Anchor-Robust optimization

How to nest anchored sets into optimization problems?

- Consider **anchored solutions** as triplets :

$(x^0,$	$H,$	$(x^\delta)_{\delta \in \Delta})$
baseline	anchored set	second-stage
solution		solutions

- **Criteria :**

Baseline cost	Anchoring weight or size	Second-stage cost
$c^T x^0$	$w(H) = \sum_{i \in H} w_i$ or $ H $	$\max_{\delta \in \Delta} c^\delta{}^T x^\delta$

Anchor-Robust optimization

How to nest anchored sets into optimization problems ?

► Consider **anchored solutions** as triplets :

$$\begin{array}{ccc} (x^0, & H, & (x^\delta)_{\delta \in \Delta}) \\ \text{baseline} & \text{anchored set} & \text{second-stage} \\ \text{solution} & & \text{solutions} \end{array}$$

► **Criteria :**

$$\begin{array}{ccc} \text{Baseline cost} & \text{Anchoring weight or size} & \text{Second-stage cost} \\ c^T x^0 & w(H) = \sum_{i \in H} w_i \text{ or } |H| & \max_{\delta \in \Delta} c^\delta{}^T x^\delta \end{array}$$

► **Anchor-Robust problems :**

Written as max/min/max problems, e.g.,

$$\begin{array}{lll} \max & \min & \max \\ x^0 \in \mathcal{X} & \delta \in \Delta & x^\delta \in \mathcal{X}^\delta \\ c^T x^0 \leq M & & x_i^0 = x_i^\delta \quad \forall i \in H \\ H \subseteq I & & \end{array} \quad w(H)$$

$= w(H)$ iff H is an anchored set

Challenges in robust optimization

► **Static-robustness** [Kouvelis Yu, 1996 ; Bertsimas Sim, 2004]

- Choice of Δ : discrete, polytope, convex...
- Complexity : harder than deterministic problem ?
- Tractability : solvable with a Mixed-Integer Programming (MIP) solver ?

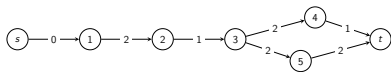
► **Adjustable/2-stage robustness** [Ben-Tal et al, 2003]

- Complexity : adjustable can be harder than static
- Tractability : challenging to solve exactly, state-of-the-art :
 - * approximation, e.g., through decision rules
[Ben-Tal et al, 2003 ; Bertsimas Goyal 2012]
 - * decomposition on exponential MIP formulations
[Billionnet et al, 2014 ; Ayoub Poss 2016]

► **Anchor-Robustness** inherits from these challenges, with additional combinatorial decisions for set H .

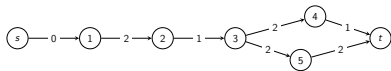
A graph model for AnchRobPSP

Precedence graph $G(p)$

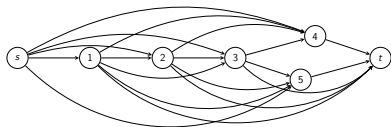


A graph model for AnchRobPSP

Precedence graph $G(p)$



G^Δ graph



Arc-weights :

$$L_{ij}^\Delta := \max_{\delta \in \Delta} \underbrace{L_{G(p+\delta)}(i,j)}_{\text{longest } i\text{-}j \text{ path in } G(p+\delta)}$$

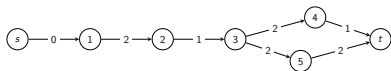
Theorem

H is anchored w.r.t. x^0

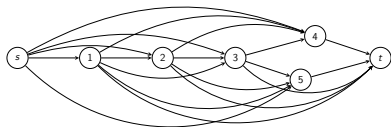
iff x^0 is a schedule of $G^\Delta[H]$, i.e.,
 $x_j^0 - x_i^0 \geq L_{ij}^\Delta$ for every $i, j \in H$

A graph model for AnchRobPSP

Precedence graph $G(p)$



G^Δ graph



Arc-weights :

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Theorem

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iff x^0 is a schedule of $G^\Delta[H]$, i.e.,

$$x_j^0 - x_i^0 \geq L_{ij}^\Delta \text{ for every } i, j \in H$$

→ Proof : from combinatorial property in rescheduling

→ Connection with L_{ij}^Δ values

Complexity landscape of AnchRobPSP

Complexity of AnchRobPSP ?

Complexity landscape of AnchRobPSP

- If L_{ij}^{Δ} values are hard to compute (e.g., Δ polytope) :
 - deciding if (x^0, H) is a feasible solution is NP-hard
 - Under $P \neq NP$, Decision-AnchRobPSP is not in NP

Complexity landscape of AnchRobPSP

► If L_{ij}^Δ values are hard to compute (e.g., Δ polytope) :

- deciding if (x^0, H) is a feasible solution is **NP-hard**
- Under $P \neq NP$, Decision-AnchRobPSP is **not in NP**

► If L_{ij}^Δ easy to compute ?

Focus on **budgeted uncertainty** : **NP-complete** v.s. **polynomial** cases

Box	polynomial		Algo $\mathbf{A}^+$
Budgeted	NP-complete	$\Gamma = 1$, G path	from Knapsack
	polynomial	any Γ , G path, unit w	Dyn. Prog.
	pseudo-polynomial	fixed Γ , G series-parallel	Dyn. Prog.
	NP-complete	$\Gamma = 1$, unit $\hat{\delta}$	from Knapsack
	polynomial	$\Gamma = 1$, unit $\hat{\delta}$, $p = 0$	Poset problem

Complexity landscape of AnchRobPSP

► If L_{ij}^{Δ} values are hard to compute (e.g., Δ polytope) :

- deciding if (x^0, H) is a feasible solution is **NP-hard**
- Under $P \neq NP$, Decision-AnchRobPSP is **not in NP**

► If L_{ij}^{Δ} easy to compute ?

Focus on **budgeted uncertainty** : **NP-complete** v.s. **polynomial** cases

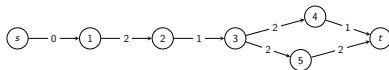
Box	polynomial		Algo $\mathbf{A}^+$
Budgeted	NP-complete	$\Gamma = 1$, G path	from Knapsack
	polynomial	any Γ , G path, unit w	Dyn. Prog.
	pseudo-polynomial	fixed Γ , G series-parallel	Dyn. Prog.
	NP-complete	$\Gamma = 1$, unit $\hat{\delta}$	from Knapsack
	polynomial	$\Gamma = 1$, unit $\hat{\delta}$, $p = 0$	Poset problem

→ Boundary between hard and polynomial cases

→ **Budgeted uncertainty is a NP-hard case**

Budgeted Δ : Layered graph

Precedence graph $G(p)$



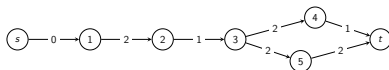
A graph model
**dedicated to budgeted
uncertainty**

Theorem

H is anchored w.r.t. x^0 iff
there exists a schedule of $G^{Lay}(H)$
that coincides with x^0 on the upper
layer.

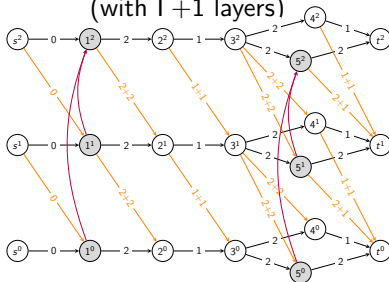
Budgeted Δ : Layered graph

Precedence graph $G(p)$



A graph model
**dedicated to budgeted
uncertainty**

Layered graph $G^{Lay}(H)$
(with $\Gamma+1$ layers)



Theorem

H is anchored w.r.t. x^0 iff
there exists a schedule of $G^{Lay}(H)$
that coincides with x^0 on the upper
layer.

Budgeted Δ : MIP formulation (Lay)

Theorem

*The layered graph yields an **MIP** formulation for (AnchRobPSP) under budgeted uncertainty.*

Budgeted Δ : MIP formulation (Lay)

Theorem

*The layered graph yields an **MIP** formulation for (AnchRobPSP) under budgeted uncertainty.*

$$\begin{aligned}
 (\text{Lay}) \quad & \max \quad \sum_{i \in J} w_i h_i \\
 \text{s.t.} \quad & \left. \begin{aligned}
 x_j^\gamma - x_i^\gamma &\geq p_i && \forall \text{ arc } (i, j), \forall \text{ layer } \gamma \\
 x_j^\gamma - x_i^{\gamma+1} &\geq p_i + \hat{\delta}_i && \forall \text{ arc } (i, j), \forall \text{ layer } \gamma < \Gamma \\
 x_j^\Gamma - x_j^\gamma &\geq -\text{big}M(1 - h_j) && \forall \text{ job } j, \forall \text{ layer } \gamma < \Gamma \\
 x_t^\Gamma &\leq M \\
 x_j^\gamma &\geq 0 && \forall \text{ job } j, \forall \text{ layer } \gamma \\
 h_j &\in \{0, 1\} && \forall \text{ job } j
 \end{aligned} \right\} \begin{array}{l} \exists \text{ schedule } x \text{ of} \\ \text{layered graph} \\ \text{satisfying} \\ \text{deadline } M \end{array} \\
 & \hspace{15em} = 1 \text{ iff } j \text{ anchored}
 \end{aligned}$$

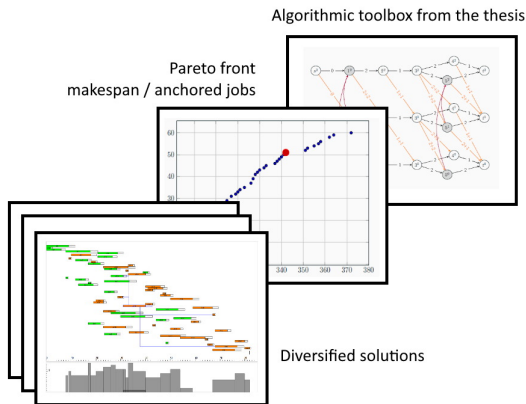
- Compact formulation (not often the case for robust 2-stage problems)
 - MIP implementation with Julia JuMP and Cplex 12.8
- Random instances with budget $\Gamma \in \{1, 2, 3\}$, up to **10 000 jobs**.
- Solutions within 1% gap found in <45 seconds.

Other results :

- ▶ MIP formulations for other uncertainty set Δ
- ▶ Polyhedral approach to improve this MIP formulations
- ▶ Extension to the NP-hard RCPSP
i.e. Project Scheduling with shared resources
 - ▶ MIP formulation
 - ▶ Efficient heuristic method
 - ▶ “Proof of Concept” software

Use case for the Anchor-Robust RCPSP

- Integration into a **decision-aiding tool**, developed at EDF in 2019



- Use case on maintenance planning for nuclear outages

- [1] The Anchor-Robust Project Scheduling Problem
P. Bendotti, P. Chrétienne, P. Fouilhoux and A. Pass-Lanneau (to appear)

- [2] Dominance-based linear formul. for the Anchor-Robust Project Scheduling Pbm.
P. Bendotti, P. Chrétienne, P. Fouilhoux and A. Pass-Lanneau
European Journal of Operational Research (EJOR) 295 :1(22-33) 2021

- [3] Anchored Rescheduling Problems under Generalized Precedence Constraints
P. Bendotti, P. Chrétienne, P. Fouilhoux and A. Pass-Lanneau
International Symposium on Combinatorial Optimization (ISCO 2020)
Conference acts in Lecture Notes in Computer Science ISCO (LNCS)
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- [4] Anchored reactive and proactive solutions to the CPM-scheduling problem
P. Bendotti, P. Chrétienne, P. Fouilhoux and A. Quilliot
European Journal of Operational Research (EJOR) 261 :1(67-74) 2017